CHMENG 250: Transport Processes

Homework 1: Mathematical Foundations

Assigned: Tue, Aug. 25, 9:00 a.m. (PT) **Due:** Sat, Sep. 5, 11:59 p.m. (PT) **Total:** 100 points

Goal: Develop fluency in symbolic and Einstein's indicial notation; manipulate the Kronecker delta and Levi–Civita symbol; prove and interpret vector and tensor identities; analyze velocity gradients, material acceleration, and fluid motion; and formulate, nondimensionalize, solve, and interpret a steady transport problem.

Convention. Unless stated otherwise, all indices range over $1, 2, 3$, repeated indices are summed, and all scalar, vector, and tensor fields are sufficiently smooth for the operations used.

Problem 1: Indicial notation [15 points]

Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{R}^3$ be vectors, let $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathrm{Lin}(\mathbb{R}^3)$ be second-rank tensors, and let $s \in \mathbb{R}$ be a scalar.

(a) For each expression, identify the free and dummy indices and classify the result as a scalar, vector, or second-rank tensor.

(i) $A_{ij}u_j$ (ii) $A_{ij}B_{ji}$ (iii) $\epsilon_{ijk}u_jv_k$ (iv) $\delta_{ij}A_{jk}$ [4 points]

(b) Determine whether each expression is well formed under Einstein's summation convention. For each malformed expression, explain the error and give one valid correction.

(i) $A_{ij}u_j = v_i$ (ii) $A_{ij}u_j = v_j$ (iii) $A_{ij} + u_i$

(iv) $A_{ij}B_{jk}u_j$ (v) $A_{ij}B_{jk}C_{ki}$ [5 points]

(c) Translate each component relation into symbolic (Gibbs) notation.

(i) $c_i = A_{ij}b_j$ (ii) $C_{ij} = A_{ik}B_{kj}$ (iii) $s = A_{ij}B_{ij}$ [6 points]

Problem 2: Kronecker delta and Levi–Civita contractions [20 points]

Let $\boldsymbol{a} \in \mathbb{R}^3$ be arbitrary.

(a) Evaluate each contraction.

(i) δ_{ii} (ii) $\delta_{ij}a_j$ (iii) $\delta_{ij}\delta_{jk}$ (iv) $\delta_{ij}\epsilon_{ijk}$ [4 points]

(b) Prove the epsilon–delta identity

$$\epsilon_{ijk}\epsilon_{i\ell m} = \delta_{j\ell}\delta_{km} - \delta_{jm}\delta_{k\ell}.$$ [8 points]

Hint

Consider separately the cases in which either ordered pair has equal entries, the pairs are identical, the pairs are reversed, or they contain different sets of indices.

(c) From Problem 2(b), deduce

$$\epsilon_{ijk}\epsilon_{ijm} = 2\delta_{km},$$
$$\epsilon_{ijk}\epsilon_{ijk} = 6.$$

[4 points]

(d) For arbitrary $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z} \in \mathbb{R}^3$, prove Lagrange's identity

$$(\boldsymbol{u} \times \boldsymbol{v}) \cdot (\boldsymbol{w} \times \boldsymbol{z}) = (\boldsymbol{u} \cdot \boldsymbol{w})(\boldsymbol{v} \cdot \boldsymbol{z}) - (\boldsymbol{u} \cdot \boldsymbol{z})(\boldsymbol{v} \cdot \boldsymbol{w}).$$

[4 points]

Problem 3: Convective acceleration and vorticity [20 points]

Let $\boldsymbol{v}(\boldsymbol{x})$ be a steady velocity field with vorticity $\boldsymbol{\omega} \coloneqq \boldsymbol{\nabla} \times \boldsymbol{v}$.

(a) Use indicial notation to prove

$$(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v} = \boldsymbol{\nabla}\left(\frac{1}{2}|\boldsymbol{v}|^2\right) - \boldsymbol{v} \times \boldsymbol{\omega}. \tag{1}$$

[10 points]

Hint

Work with $(\boldsymbol{v} \times \boldsymbol{\omega})_i$ and use the epsilon–delta identity from Problem 2(b).

(b) Consider the simple shear

$$\boldsymbol{v}(\boldsymbol{x}) = \dot{\gamma} x_2 \boldsymbol{e}_1,$$

with constant shear rate $\dot{\gamma}$. Calculate $\boldsymbol{\omega}$ and the two terms on the right-hand side of Eq. (1), and verify that the convective acceleration vanishes. [5 points]

(c) Consider solid-body rotation about the x_3 axis,

$$\boldsymbol{v}(\boldsymbol{x}) = \Omega\,(-x_2\boldsymbol{e}_1 + x_1\boldsymbol{e}_2),$$

with constant angular speed Ω. Define $\boldsymbol{x}_\perp \coloneqq x_1\boldsymbol{e}_1 + x_2\boldsymbol{e}_2$, $r \coloneqq \sqrt{x_1^2 + x_2^2}$, and $\boldsymbol{e}_r \coloneqq \boldsymbol{x}_\perp / r$ for $r > 0$. Use Eq. (1) to show that

$$(\boldsymbol{\nabla}\boldsymbol{v})\,\boldsymbol{v} = -\Omega^2 \boldsymbol{x}_\perp = -\Omega^2 r \boldsymbol{e}_r,$$

and explain why the acceleration is centripetal. [5 points]

Problem 4: Anatomy of a flow [20 points]

Consider the steady velocity field

$$\boldsymbol{v}(\boldsymbol{x}) = \gamma x_2 \boldsymbol{e}_1 + \beta x_1 \boldsymbol{e}_2,$$

where β and γ are constants with dimensions of inverse time.

(a) Compute the velocity gradient $\boldsymbol{L} \coloneqq \boldsymbol{\nabla} \boldsymbol{v}$, divergence $\boldsymbol{\nabla} \cdot \boldsymbol{v}$, and vorticity $\boldsymbol{\omega} \coloneqq \boldsymbol{\nabla} \times \boldsymbol{v}$. [5 points]

(b) Compute the rate-of-strain tensor $\boldsymbol{D}$ and rotation tensor $\boldsymbol{W}$. Determine the conditions on β and γ for pure strain ($\boldsymbol{W} = \boldsymbol{0}$, equivalently $\boldsymbol{\omega} = \boldsymbol{0}$) and for solid-body rotation ($\boldsymbol{D} = \boldsymbol{0}$). [5 points]

(c) Compute the material acceleration $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$ and explain why it need not vanish in a steady flow. [4 points]

(d) Write the pathline equations $\dot{\boldsymbol{x}} = \boldsymbol{v}(\boldsymbol{x})$ and derive second-order ODEs for $x_1(t)$ and $x_2(t)$. Describe the pathlines for $\beta\gamma < 0$, $\beta\gamma > 0$, and $\beta\gamma = 0$. [6 points]

Problem 5: Steady heat conduction with generation [25 points]

A planar solid occupies $0 \le x \le L$ and has constant thermal conductivity k and uniform volumetric heat-generation rate $\dot{q}_v > 0$. The surface at $x = 0$ is insulated, while the surface at $x = L$ loses heat by convection to an ambient fluid at temperature T_∞ with heat-transfer coefficient h. Assume steady one-dimensional conduction and neglect radiation.

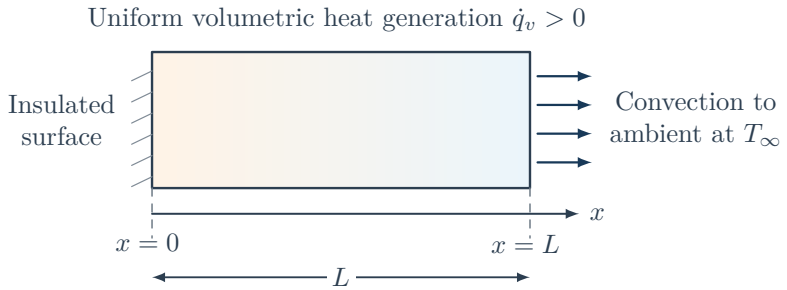


Figure 1: Steady one-dimensional conduction in a slab with uniform volumetric heat generation, insulation at $x = 0$, and convection to an ambient fluid at T_∞ at $x = L$. The shading is schematic.

(a) Apply a steady energy balance to a differential slab to derive the differential equation for $q_x(x)$. Using Fourier's law $\boldsymbol{q} = -k\boldsymbol{\nabla} T$, formulate the boundary-value problem for $T(x)$. [6 points]

> **Hint**
> Balance a slab from x to $x + \mathrm{d}x$ with unit cross-sectional area.

(b) Solve the boundary-value problem for $T(x)$. [5 points]

(c) Determine $q_x(x)$ and verify the global steady energy balance per unit area

$$\int_0^L \dot{q}_v \, \mathrm{d}x = q_x(L) = h\,[T(L) - T_\infty]\,. \qquad \text{[4 points]}$$

(d) Define

$$X \coloneqq \frac{x}{L}, \quad \theta \coloneqq \frac{k\,(T - T_\infty)}{\dot{q}_v L^2}, \quad \mathrm{Bi} \coloneqq \frac{hL}{k}.$$

The Biot number is the ratio of the internal conductive resistance scale L/k to the external convective resistance scale $1/h$. Nondimensionalize the boundary-value problem and solve for $\theta(X)$. [5 points]

(e) Locate the maximum temperature and calculate its value. Interpret the limits $\mathrm{Bi} \to \infty$ and $\mathrm{Bi} \to 0^+$, and explain why no finite steady solution exists at $\mathrm{Bi} = 0$ when $\dot{q}_v > 0$. [5 points]